A Hidden Ocean Wave Is Raising the Sea on the West Coast

From: NPR World Mined: 2026-10-20 3 min read

A giant wave you cannot see is raising **sea levels** on the West Coast of the United States. It is called a **Kelvin wave**. Scientists say more are coming.

A **Kelvin wave** is not a wave you could surf. It moves slowly under the ocean's surface and stretches for thousands of miles. Scientist Dillon Amaya called it "a **planetary-scale** wave."

These waves can form during **El Niño** years, when a big part of the Pacific Ocean gets much warmer than usual. The warm water travels east under the sea to South America. Then part of it moves north along the West Coast.

The wave does not cause floods by itself. But it makes floods more likely when big storms come. In early October, **sea levels** in parts of Southern California were about six to eight inches higher than normal. Then Tropical Storm Rachel flooded some neighborhoods.

The warmer water also brings surprises. A **whale shark** was spotted off San Diego. The next **Kelvin wave** may reach San Diego in late October or early November.

Word Treasure

Kelvin wave -- A huge, slow wave under the ocean that raises the sea along coasts.

sea levels -- How high the surface of the ocean is.

El Niño -- A time when part of the Pacific Ocean gets much warmer than usual.

planetary-scale -- So big that it covers a large part of the planet.

whale shark -- The biggest fish in the world, which is gentle and eats tiny sea creatures.

Background you need

The ocean is always moving, and some of its waves travel deep under the surface where we cannot see them.

Warm water moving along a coast can make the sea there rise a little.

When the sea is already higher, a big storm can push water onto land more easily.

Break it down

WHO: Scientists, including Dillon Amaya, who study the ocean
WHAT: A hidden Kelvin wave is raising sea levels
WHERE: The West Coast of the United States, including Southern California and San Diego
WHY: Warm water from El Niño moved under the sea and up the coast

Why it matters

Knowing when the sea will be higher helps families who live near the coast get ready for floods.

Test what you remember

1. What is raising sea levels on the West Coast?

- (A) A big ship
- (B) A Kelvin wave
- (C) Melting snowmen
- (D) A volcano

2. Can you surf a Kelvin wave?

- (A) Only at night
- (B) Yes, it is a great surfing wave
- (C) Only in winter
- (D) No, it moves slowly under the ocean's surface

3. When can these waves form?

- (A) Only on weekends
- (B) During El Niño years
- (C) When the Moon is full
- (D) When it snows

4. How much higher were sea levels in parts of Southern California in early October?

- (A) Not higher at all
- (B) About six to eight inches
- (C) About one mile
- (D) About ten feet

5. Which storm flooded some neighborhoods?

- (A) Blizzard Bob
- (B) Storm Kelvin
- (C) Hurricane Max
- (D) Tropical Storm Rachel

6. What big fish was spotted off San Diego?

- (A) A swordfish
- (B) A goldfish
- (C) A whale shark
- (D) A catfish

Answer key (try the quiz first!): 1: B 2: D 3: B 4: B 5: D 6: C

Questions to think about

1. Dillon Amaya: Calls the Kelvin wave "a planetary-scale wave."

MY THOUGHTS (at least 20 words):